

MC pileup profile -- does one profile serve signal and QCD?

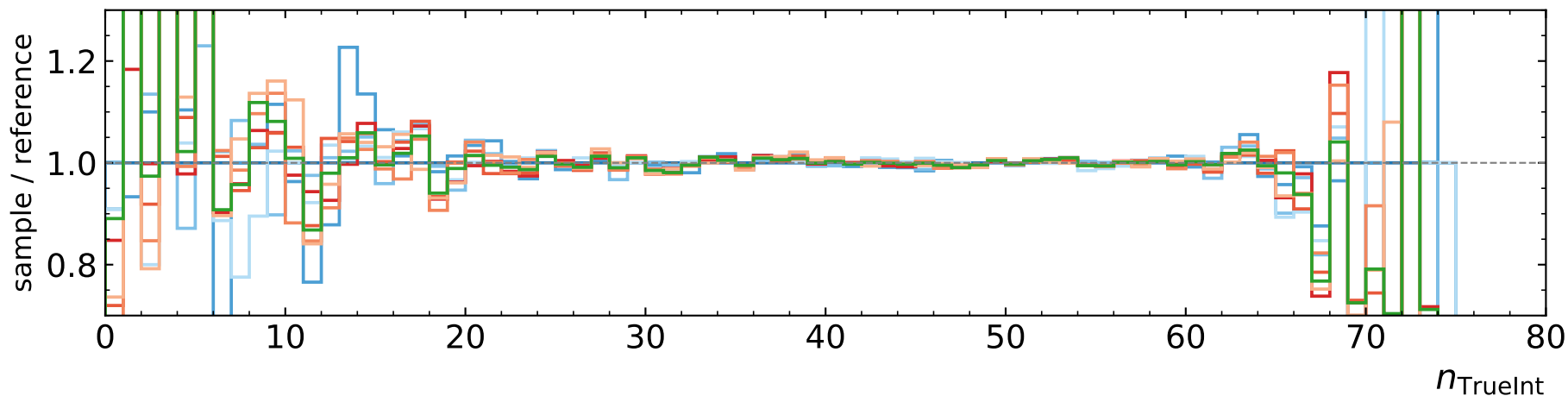
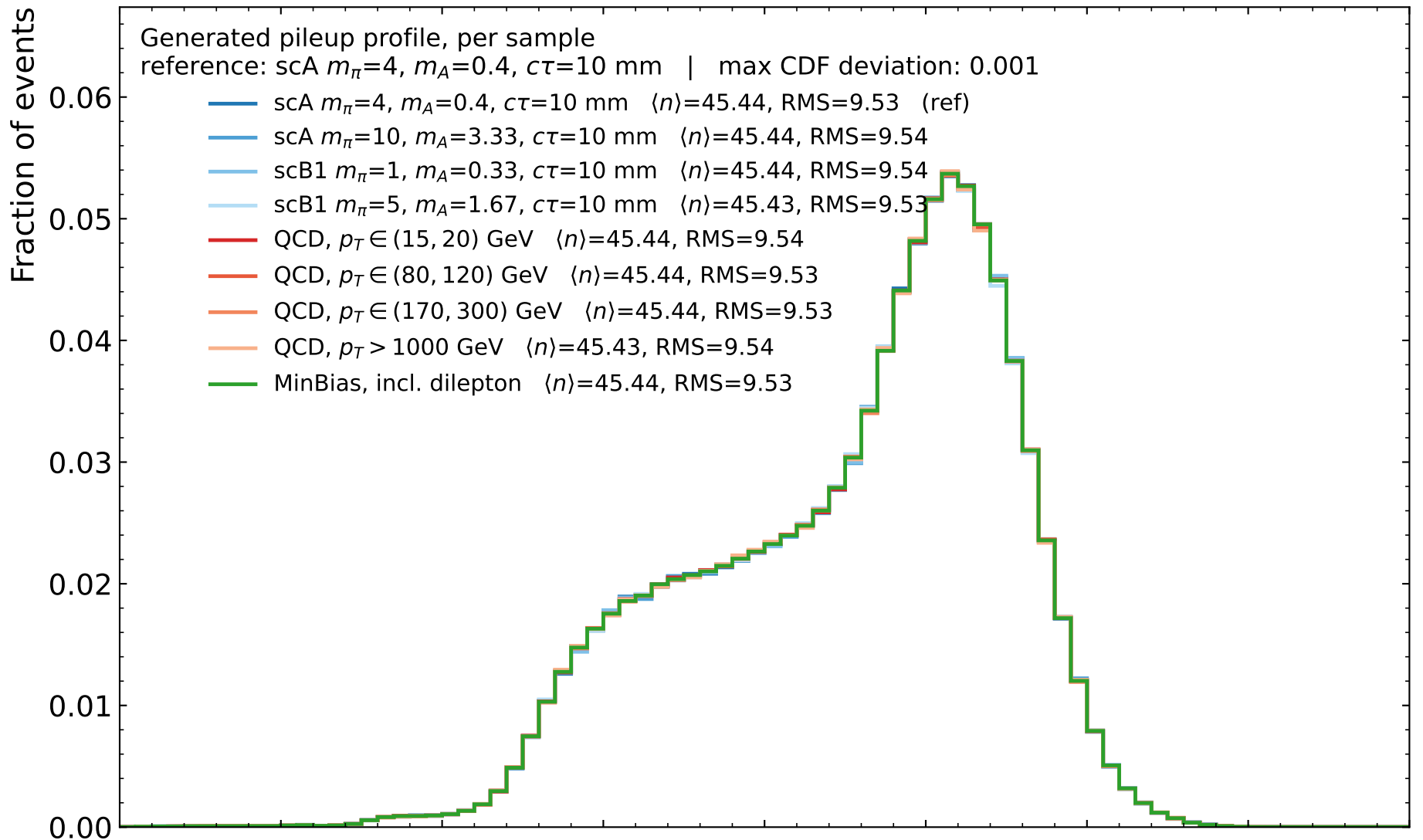
CMS *Simulation Preliminary*

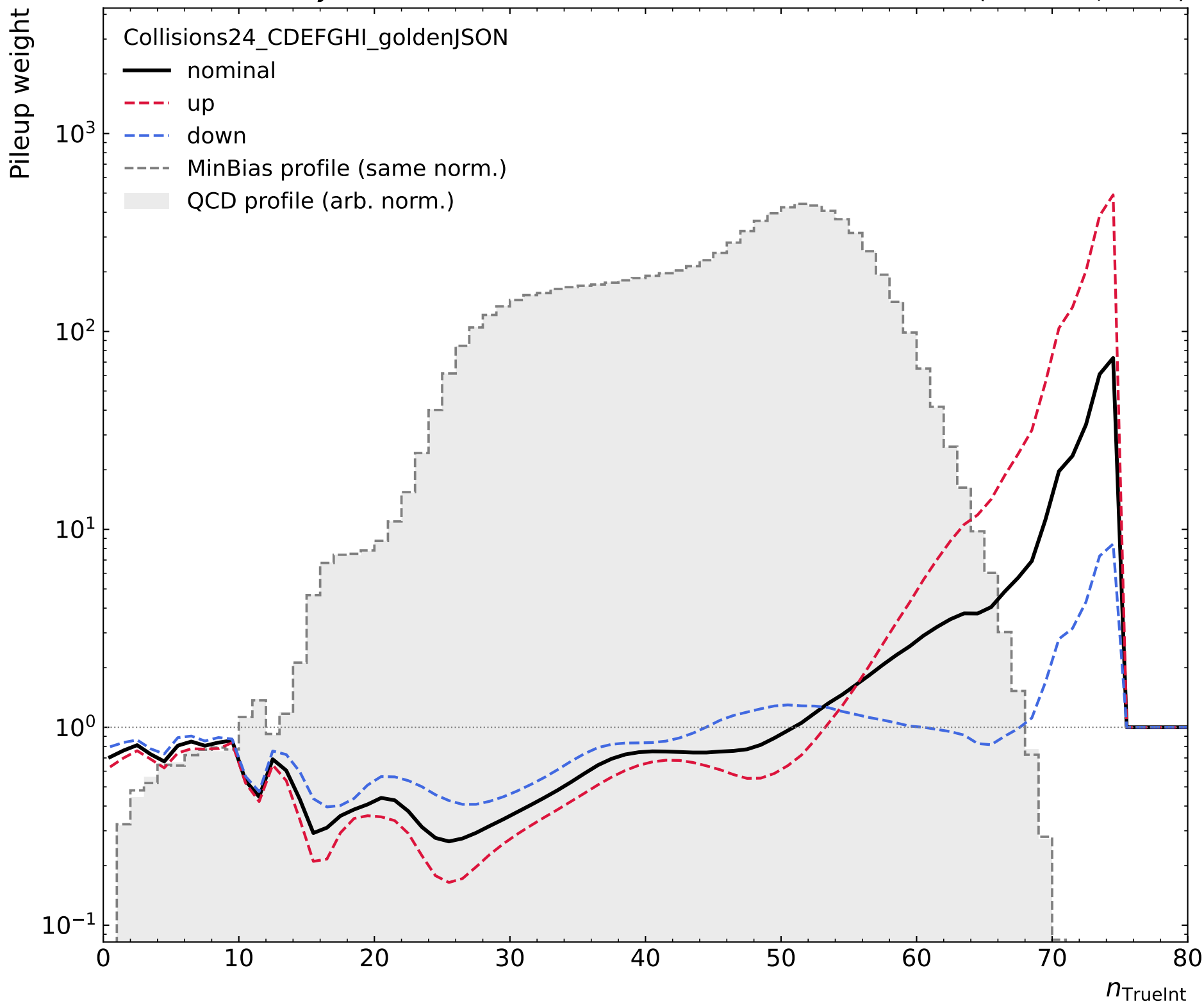
(13.6 TeV, 2024)

Generated pileup profile, per sample

reference: scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm | max CDF deviation: 0.001

- scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm $\langle n \rangle=45.44$, RMS=9.53 (ref)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm $\langle n \rangle=45.44$, RMS=9.54
- scB1 $m_\pi=1$, $m_A=0.33$, $c\tau=10$ mm $\langle n \rangle=45.44$, RMS=9.54
- scB1 $m_\pi=5$, $m_A=1.67$, $c\tau=10$ mm $\langle n \rangle=45.43$, RMS=9.53
- QCD, $p_T \in (15, 20)$ GeV $\langle n \rangle=45.44$, RMS=9.54
- QCD, $p_T \in (80, 120)$ GeV $\langle n \rangle=45.44$, RMS=9.53
- QCD, $p_T \in (170, 300)$ GeV $\langle n \rangle=45.44$, RMS=9.53
- QCD, $p_T > 1000$ GeV $\langle n \rangle=45.43$, RMS=9.54
- MinBias, incl. dilepton $\langle n \rangle=45.44$, RMS=9.53

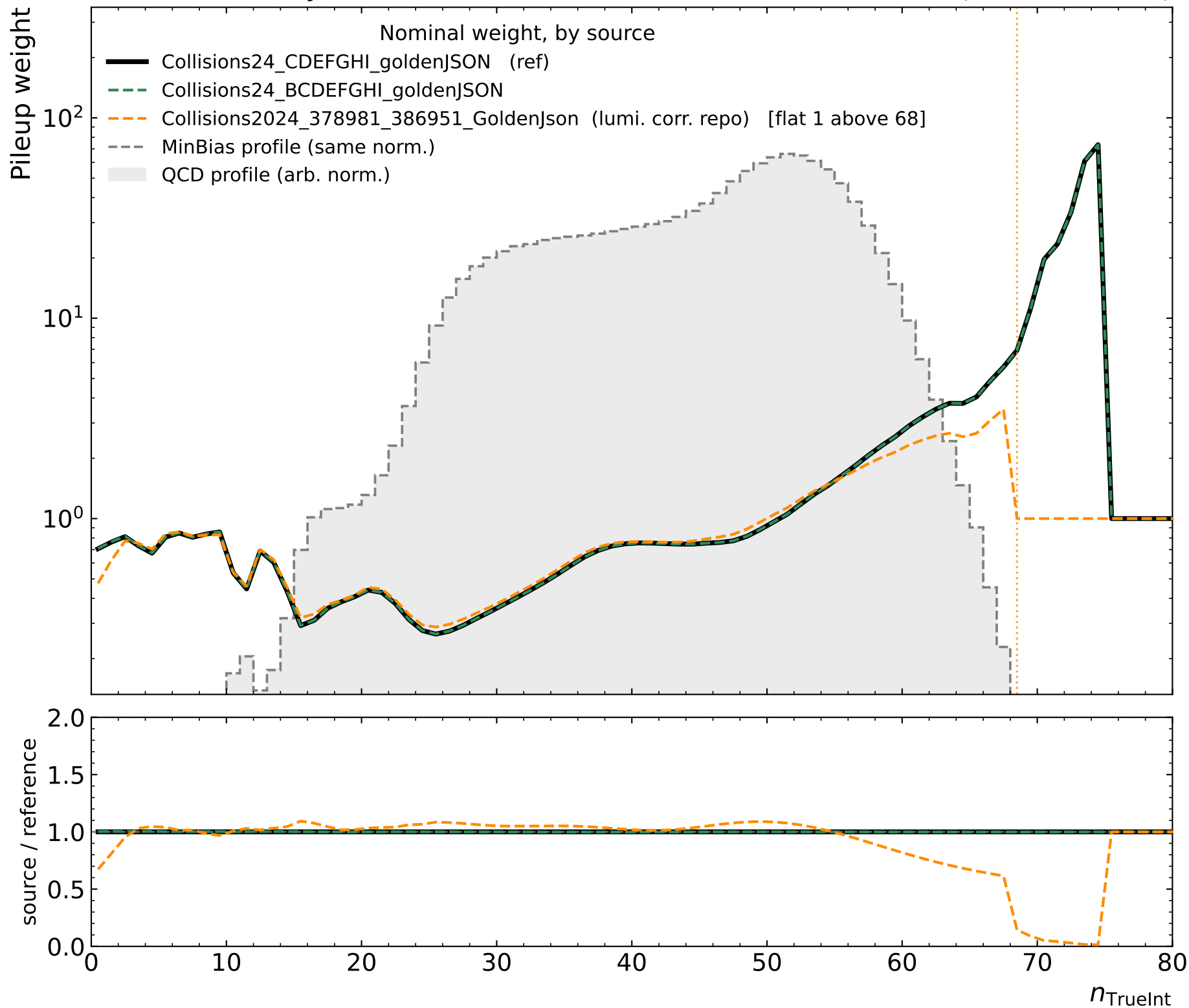




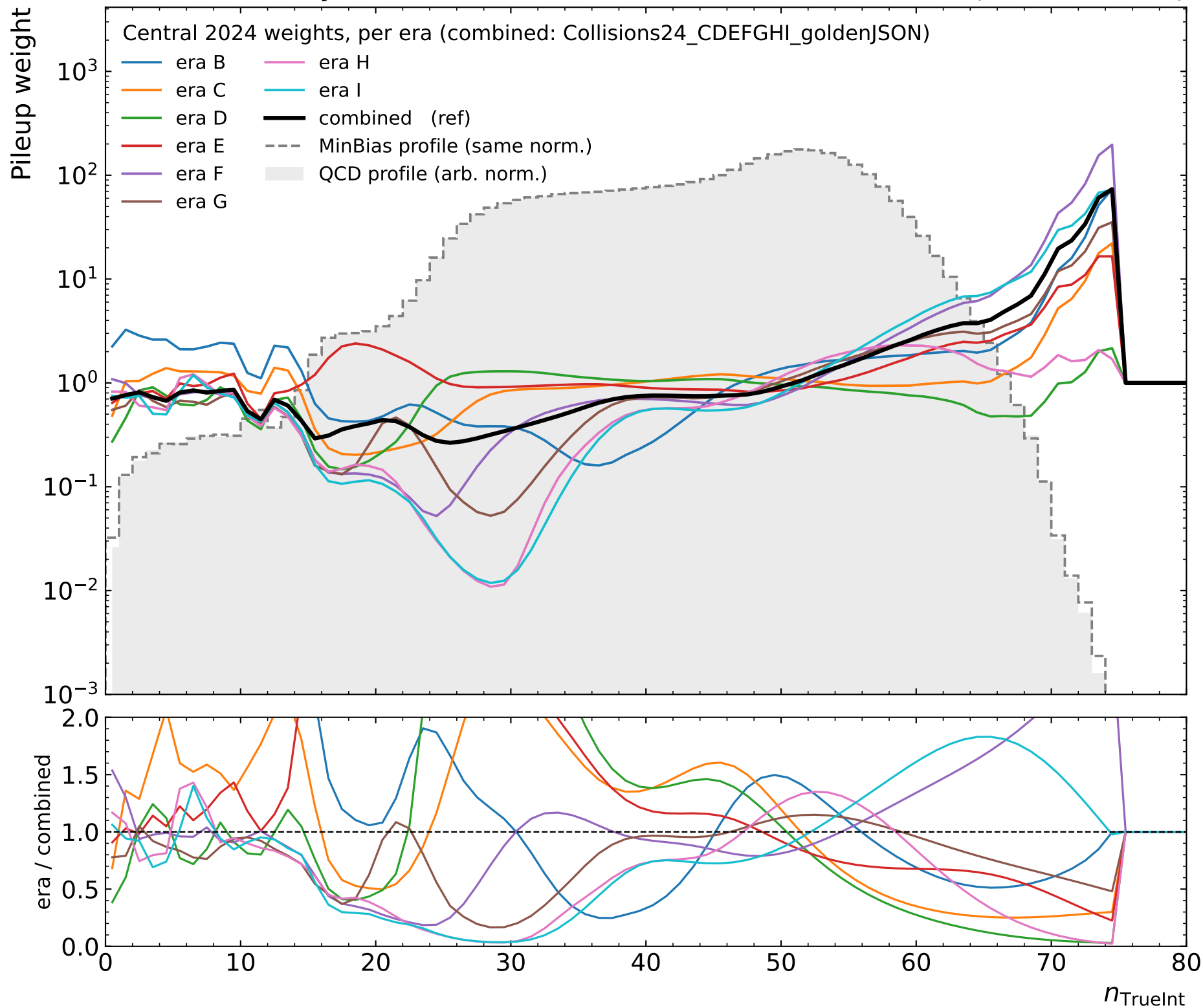
Weight sources compared -- central vs the lum-corrections stopgap

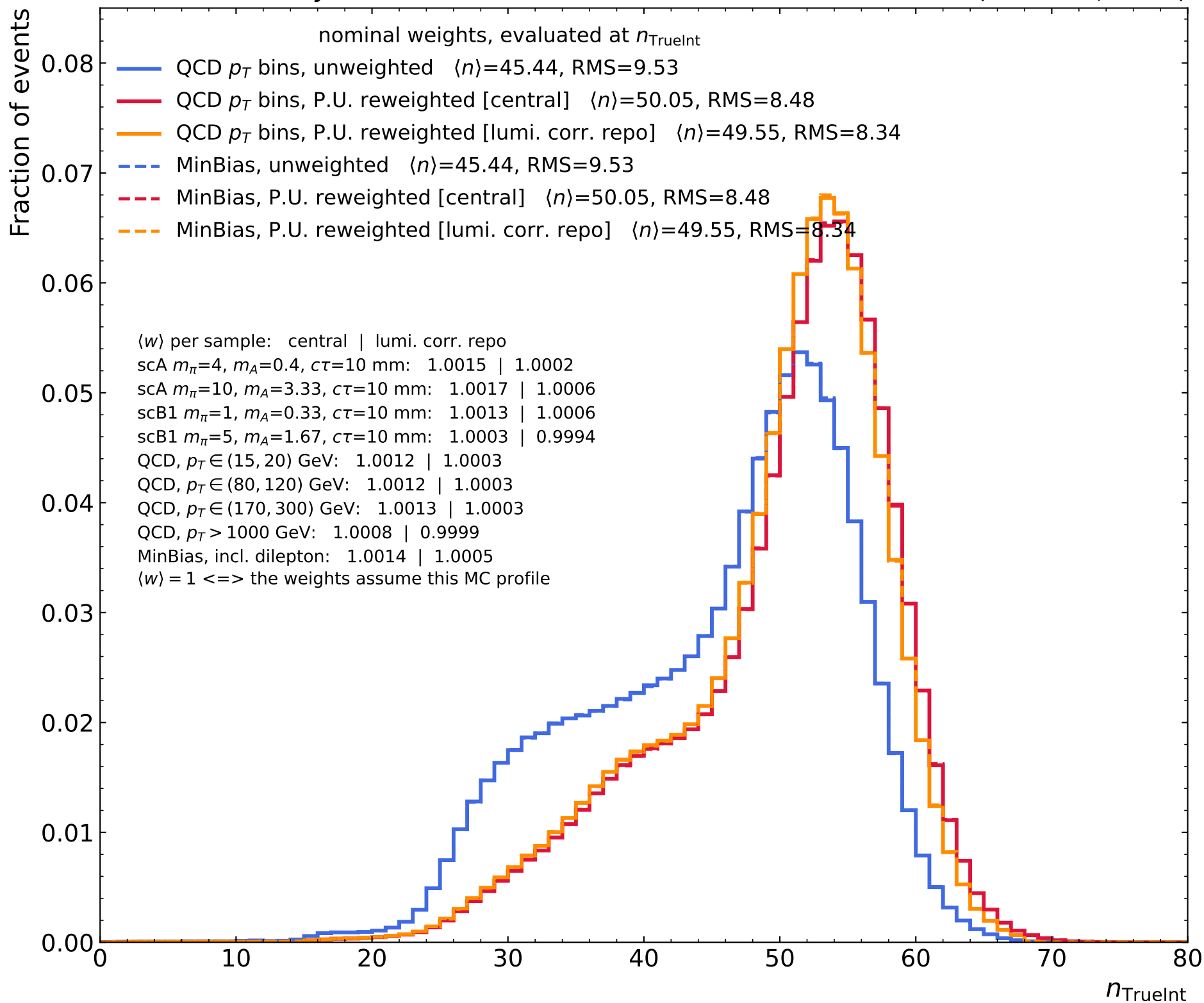
CMS Preliminary

(13.6 TeV, 2024)

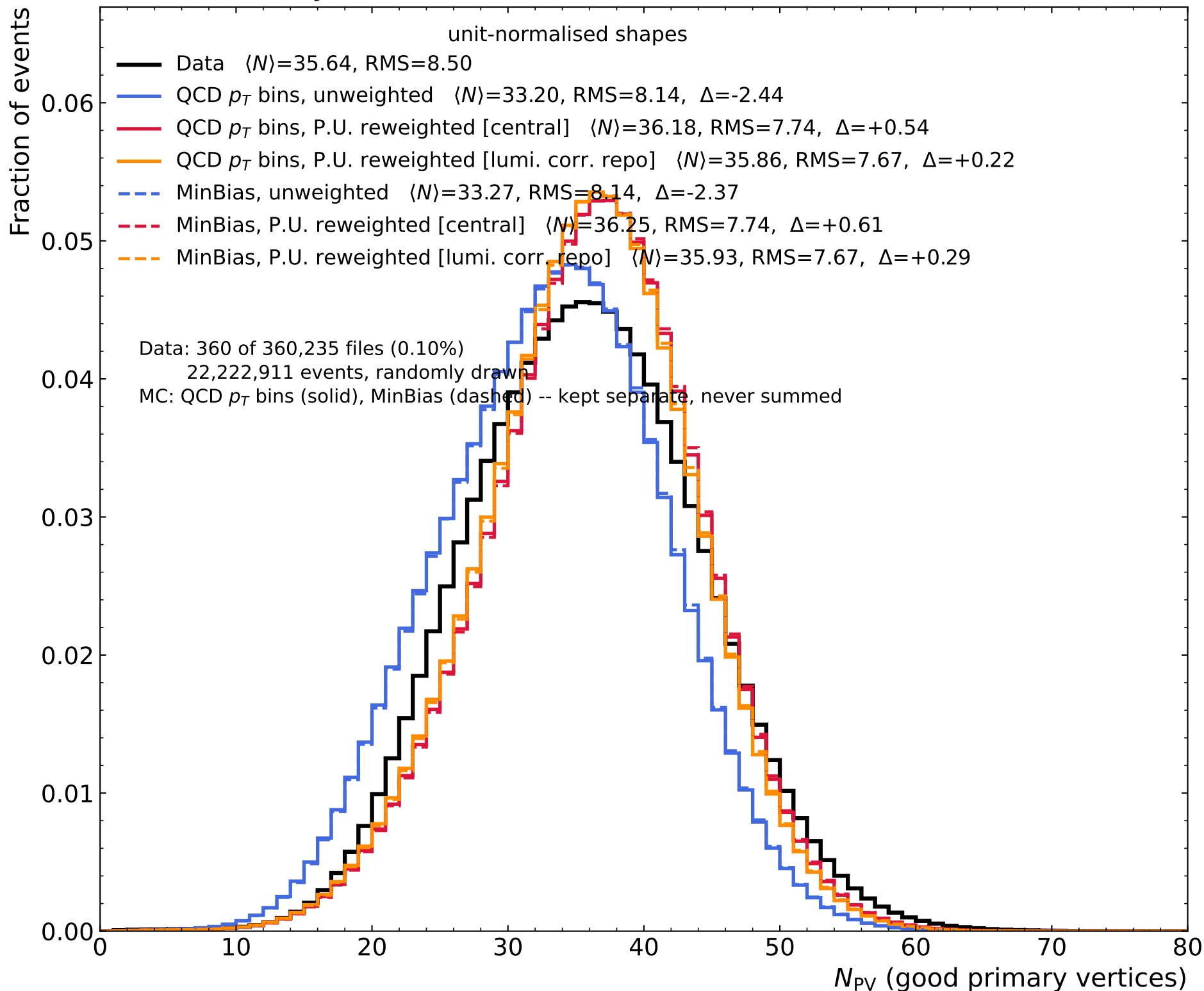


Per-era pileup weights -- the handle a single merged histogram cannot give
CMS Preliminary (13.6 TeV, 2024)





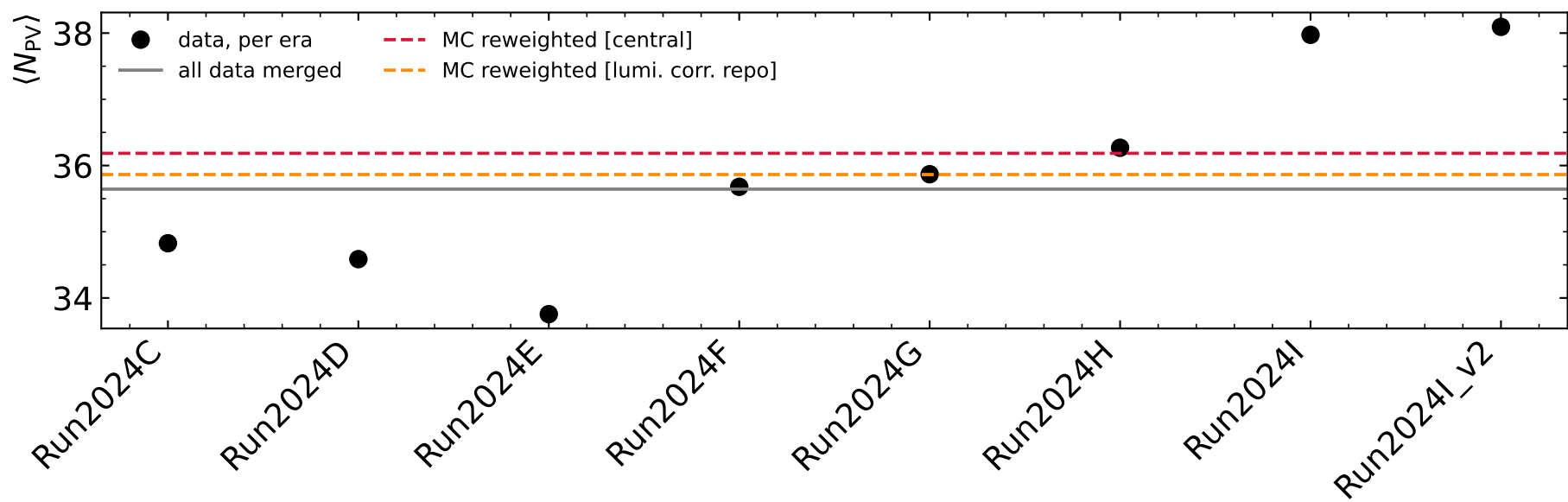
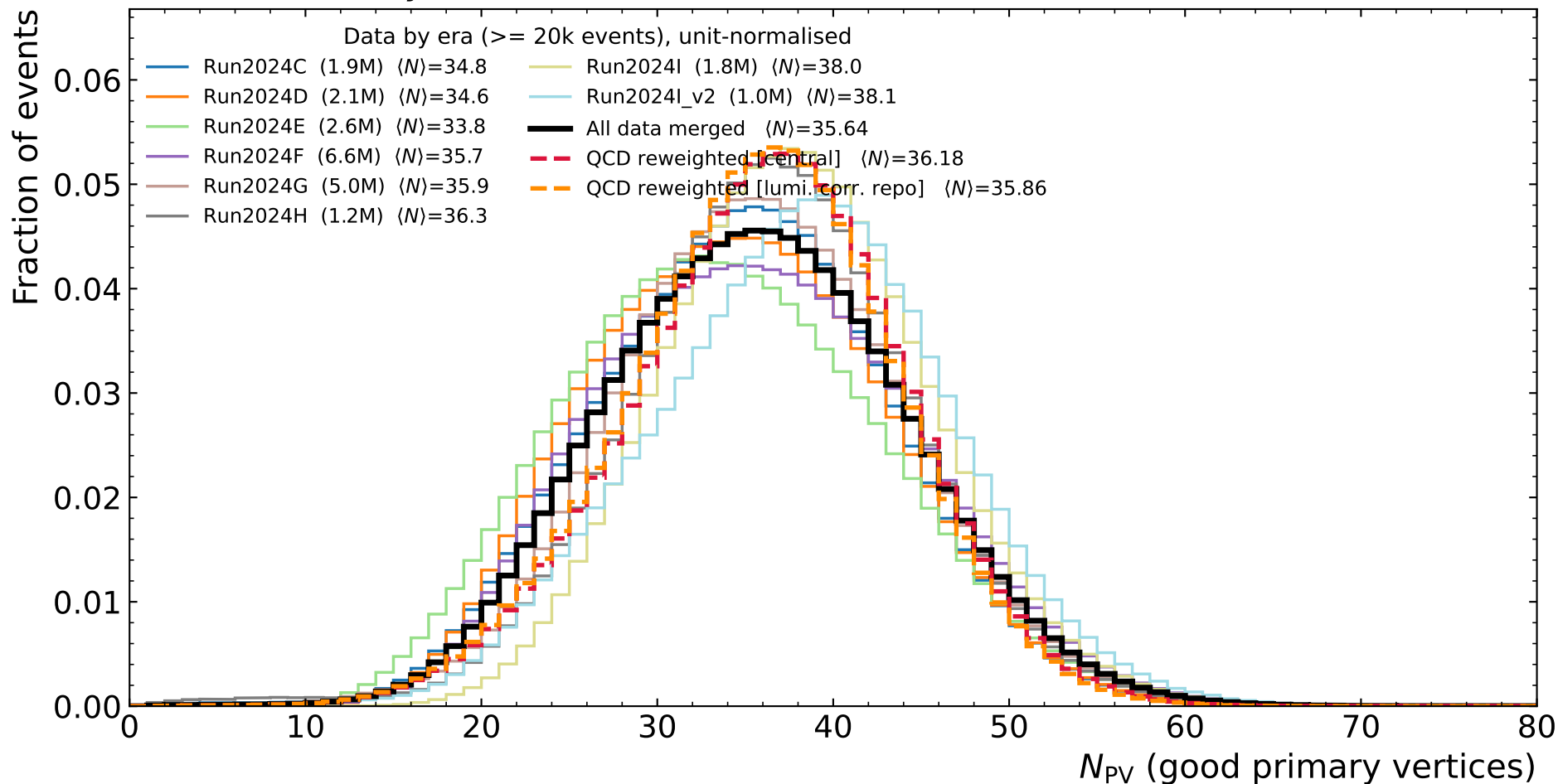
Data validation -- does the pileup correction improve the agreement?

CMS Preliminary0.11 fb⁻¹ (13.6 TeV, 2024)

Data pileup by era -- is the merged data/MC offset an averaging effect?

CMS Preliminary

0.11 fb⁻¹ (13.6 TeV, 2024)



Era-matched validation -- data vs MC weighted with that era's own correction

